AI research execution controls, 18 October 2026

These are actual bounded, CPU-only controls for audit items 8, 10, 11 and 12. They produced no new index-calculus speedup, safe learned rejection rule, or resolution of an open problem. The useful outcomes are a negative branching comparison, detection of empty-base training failure, an exact finite bucket signal, and a degree surrogate whose calibration limits are explicit.

All calculations used one worker per script with no GPU and F4_F2_ECHELON=host. Container timings are unqualified exploratory CPU diagnostics. They are not host-isolated performance evidence. No complete IC pipeline, same-target rho comparison, or deployed-curve attack ran here.

Item 8: real WDSat branching hook, negative result

sat_controls.py executed **88 solver calls**: 48 training calls selecting among six static coefficient orders, followed by 40 heldout calls comparing five policies on the same eight targets. It used the upstream WDSat commit 61c6ff3f49445af2729274819edb27b85a89efc1, the actual -g branch-order option, and the symmetric S4 model of degree at most three. Training used (n,ell,b)=(11,3,1),(11,3,2); heldout cells were (13,3,3),(17,4,5). Seeds and every ANF input are retained.

The learned policy selected the ordinary ascending coefficient order. On the eight heldout targets, total conflicts were:

Policy	Conflicts	Verified answers	Timeouts/errors
Paper default with symmetry	3,092	8/8	0/0
Block core, symmetry disabled	16,097	8/8	0/0
Interleaved core, symmetry disabled	16,785	8/8	0/0
Fixed random core, symmetry disabled	16,676	8/8	0/0
Learned static core, symmetry disabled	16,097	8/8	0/0

Every SAT assignment was replayed through the original elliptic-curve group law; every SAT/UNSAT answer agreed with complete small-base group-law membership enumeration. This is not an UNSAT proof-trace certificate. The controls disable symmetry for arbitrary orders because the upstream symmetry code interprets specific relative positions in the order. That accounting matters: the learned policy did not improve the ordinary block order and was worse than the paper control.

The search charged 0.209834 child CPU seconds for training solver calls and 0.743986 parent CPU seconds for training orchestration. Compiler/model setup, individual CPU/wall values, raw output and workload hashes remain separate in sat-setup.json, sat-raw.jsonl, sat-targets.json and sat-results.json. The initial GCC14 type error was retained. A declaration/definition/local pointer used only as memcpy source was corrected to void* in an owned copy, as specified by SAT-BUILD-PATCH.json; upstream was not edited. Learned neural branching, XOR-CDCL, cubing, cube coverage and an n41 ladder remain open.

Item 11a: exact x-coordinate labels expose an unusable training base

prefilter_search.py counted full toy curves, admitted an exact odd prime-order subgroup, enumerated 27 cells and **7,254 nonidentity subgroup targets**, and generated exhaustive three-sum labels. Each positive target has a group-law-replayed witness. An independent scalar sumset dynamic program checked the entire support, including negative labels. Both signs,

repeated summands, actual usable-base cardinality and all ordered tuple multiplicities are explicit in prefilter-inputs.jsonl.

The frozen training cell n7,b1,ell3,multiplier1 has **zero usable subgroup points** and 28 negative targets. Therefore both trained models are one-class models. At probability threshold 0.25 they reject every positive heldout:

Split	Targets	Decomposable targets	False negatives per learned model
Calibration base	28	4	4
New curves	588	48	48
New base	28	4	4
New field size	6,582	8	8

The all-accept baseline has zero false negatives and no rejections. The models are never wired into a decomposition oracle. This is an unsafe-transfer diagnostic, not evidence for a necessary condition or a new invariant. The heldouts do not rescue the training cell retrospectively.

The first attempt failed the odd-prime subgroup gate; the second found an unhandled one-class fit. Both failures and the subsequent explicit amendments are retained. In particular n7,b2 and n11,b2 have 2-power group orders, and n11,b1 has a repeated largest prime factor and was excluded by the frozen simple-subgroup admission rule. A useful nondegenerate learning experiment with genuinely separated curves, bases and larger sizes is still owed.

Item 11b: the actual single-bucket question

bucket_gain.py separately measured the statistic requested by crypto-autoresearcher:knowledge/open-problems/KN-OPEN-91d4f6.md: G=M_eff*max_(s,c) Pr[h(Q+R)=c | h(Q) XOR h(R)=s]. The offset is s XOR c. This is a maximum over conditional buckets, not average agreement.

The exact subgroups have orders 29 at n7 and 2,003 at n13. The maps are F2-linear x-functionals with j=1..4, alongside an x-only SHA-derived null matched to the effective image and the same ambient XOR alphabet. Nonempty images need not have power-of-two size on a finite subgroup; the first failed assumption and its correction are retained in BUCKET-AMENDMENT-01.json.

At n13 the strongest of the tested linear maps has **G=1.4375**, while its matched null has **G=1.0817**. That is a measured finite level-one signal. Two tiny subgroups do not establish growing gain or an asymptotic attack. At n7 finite-image effects are large and the null can exceed the linear map.

The halving trace is constant on both odd-order subgroups and has effective gain **1**. Exact full-group controls give trace gain **2** and a point-based quotient-of-order-four instrument gain **4**. The latter uses the full toy group's cyclic table and is not proposed as an efficient x-only filter. All pair totals and FFT rounding residuals were checked; n7 results were independently checked by direct pair enumeration. Exact numerator and denominator tables are in bucket-results.json. This measurement leaves the open problem open.

Item 10: restricted program search returns the incumbent

The first restricted scalar-coordinate search preserved 17 candidate/control records but encountered empty-base degeneracy. That is retained in program-results.json. It cannot show useful progress.

A separately frozen experiment, PROGRAM-02-PROTOCOL.json, searches actual group-law query kernels. Targets are supplied only as points; scalar labels are confined to an independent correctness oracle. The grammar contains ordered exhaustive triples, a pair-sum hash table, and sorted pair sums, combined with five scalar-multiplied subspace predicates. Both signs are explicitly cofactor-projected, identities discarded, and duplicates removed. Projection/setup/query operations and every candidate's search cost are charged.

After an explicitly retained empty-base admission at ell3, the amended ell4 training run at n7,b1 selected **multiplier1 with the existing pair-hash kernel** from 15 candidates, matching the baseline's six usable points. Its fitness was 288 counted group additions/probes/comparisons, including projection. The heldout curve n13,b7 was constructed only after selection. It has subgroup order 53 and 16 projected usable points. Both the selected program and incumbent took 820 counted units, decomposing all 52 nonidentity targets. Thus the search found **no new improvement**.

All target answers agreed with exhaustive triple-sum membership and positive witness replay. Across search and holdout the charged candidate work was 39,587 units; the run used 2.149112 CPU seconds. Multiplicity totals equal B^3 for every base. Different base supports are reported separately and do not constitute a free yield improvement. See program-group-results.json and program-group-inputs.json.

This is restricted enumerative program search, not an AlphaEvolve/LLM loop. Complete IC cost, calibrated unit sensitivities, compiled kernel mutation, strong rho and independent pipeline holdouts remain owed.

Item 12: fit the real archive, respect calibration units

surrogate.py admitted **1,718** unique pdp-degree-profile/1 aggregate-cell receipts from nine exact Koblitz curves (n=13,19,23,29,31,37,41,43,47). It retained source-file hashes, line numbers, workload/base/source identities and all four censored/incomplete cells. It excluded 36 collection-run rows with a different semantic schema. There were no duplicate or conflicting run IDs in this source set.

The response is completed-cell maximum recorded MutantXL D_solve, not Boolean F4 solving degree. Counted cost is a separate response. Censored observations retain their observed lower bounds and are excluded from exact response fitting; they remain in a completion classifier. That classifier's heldout accuracy merely ties the majority baseline and falsely predicts two incomplete cells complete. Its apparent 99.47% accuracy is unhelpful.

Shape-group holdout used 1,017 training, 322 complete calibration and 375 complete test cells. The fixed random forest's degree MAE was 0.07048 and descriptive 90% split-interval coverage was 89.6%; the median baseline's MAE was 1.376. Counted-cost log2 MAE was 0.39583. These are observations on a selected, heterogeneous historical archive, not a coverage theorem.

The stricter surrogate_curve.py keeps whole exact curves out of training. Its 383 test cells on n23/n31 give degree MAE **0.10041** and counted-cost log2 MAE **0.63061**. Only three independent calibration curves exist in that split. A 90% group-conformal quantile requests rank four, so the correct interval is **unbounded**. It cannot price an expensive future run with a finite calibrated upper bound. This limitation is in surrogate-curve-results.json.

Per-cell matrix memory is not fitted from cumulative main-process peak RSS. Unseen chained presentations and the proposed n23/ell6 or n29/ell7 m4 cells are rejected as unsupported. The archive has fixed orders and is not a complete presentation/order surface. Individual historical target receipts and executable historical snapshots were not reconstructed, so the full receipt-promotion gate remains open. Further independent curves, actual F4 degree/memory receipts and prospective validation are required.

Artifacts, reproducibility and remaining status

PROTOCOL.json and the individually named amendments were written before their corresponding executions. Launch receipts bind exact sources, inputs and seeds. All raw successes, failures and rejected admissions remain append-only in this directory. Executed commands use the following prefix from repository root:

F4_F2_ECHELON=host PYTHONDONTWRITEBYTECODE=1 OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 experiments/ideas_learning_20261018/

The script suffixes were prefilter_search.py, sat_controls.py, bucket_gain.py, program_group.py, surrogate.py and surrogate_curve.py; complete argv and stdout/stderr are retained. Scripts refuse to overwrite final result files. Reproduction should copy protocols and scripts to a fresh sibling directory; WDSat requires its pinned source checkout.

controls.svg is generated from editable report_visuals.py; evidence-flow.svg comes from evidence-flow.dot. This report and visuals are included in REPORT.pdf. verify.json records the independent metadata/semantic consistency checks.

Exact local curve records and full UIDs are exported in curve-records.json. They bind different generators and representations from historical aliases. No isogeny maps or canonical IC leaderboard facts changed. Canonical candidate/volcano graph sources were checked; no graph correction is implied by these stage controls. Registry integration of new exact curve records is an obligation for the integrating agent, because this task owns only the present experiment directory.

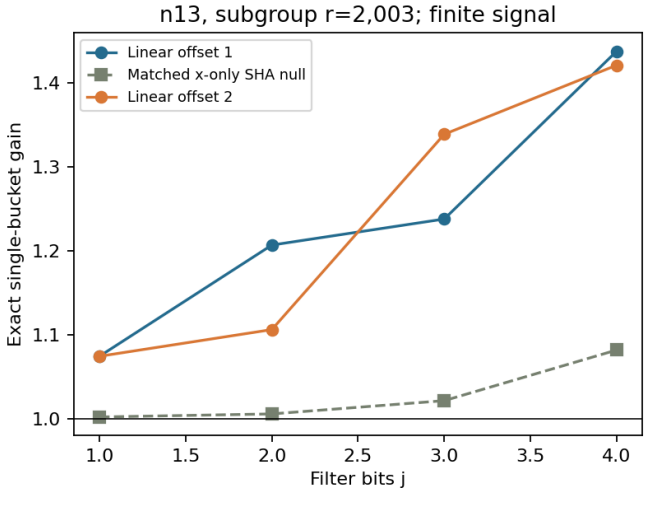
The original four audit items remain open to their full acceptance criteria. No autoresearcher experiment status or knowledge finding was promoted, and no publication submission occurred.

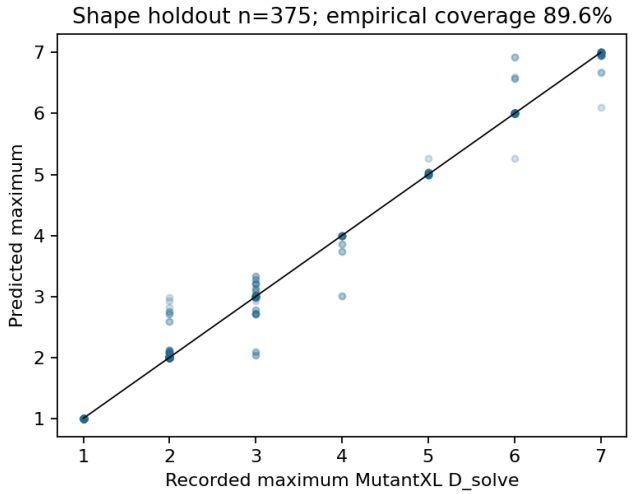
Recorded controls

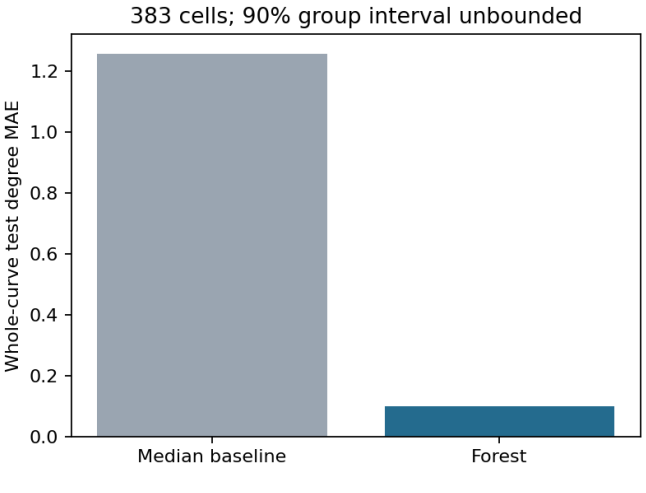
CPU-only research controls: no complete IC or security claim

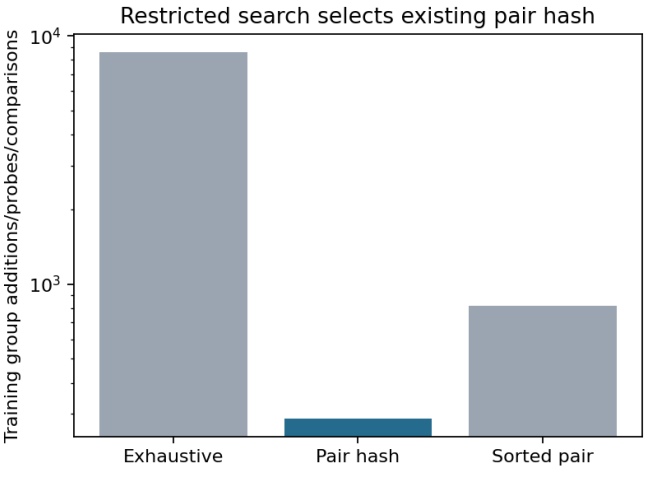


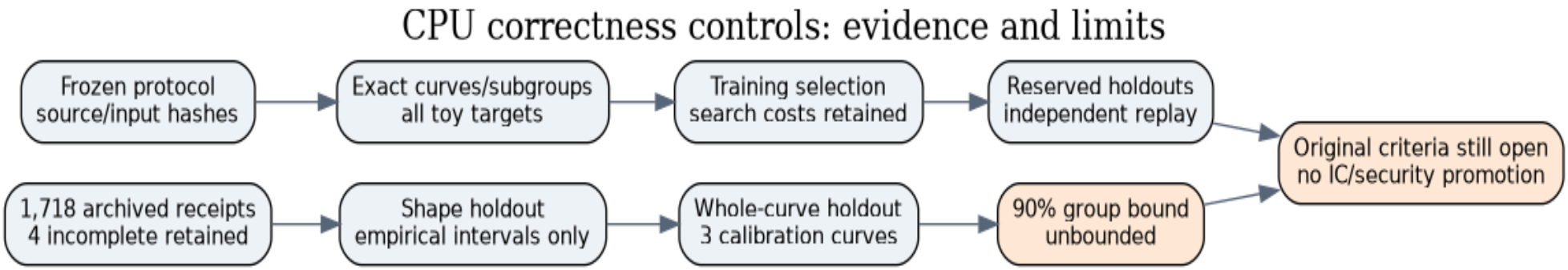











CPU correctness controls: evidence and limits
Frozen protocol
source/input hashes
Exact curves/subgroups
all toy targets
Training selection
search costs retained
Reserved holdouts
independent replay
1,718 archived receipts
4 incomplete retained
Shape holdout
empirical intervals only
Whole-curve holdout
3 calibration curves
90% group bound
unbounded
Original criteria still open
no IC/security promotion